

VIGNON Pierre-Antoine

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SUMMARY

PhD student in planetary sciences specialized in icy surface processes. My work focuses on the quantitative characterization of planetary surfaces and their behaviour using remote sensing methods; through different wavelength domains.

EDUCATION

- **PhD in Planetary Sciences** 09/29
Laboratoire de Planétologie et Géosciences France
 - In progress.
 - Large-scale study of dark spots through multi-instrumental data using computer vision, statistical analysis and atmospheric modeling methods.
- **Master's in Earth and Planetary Sciences** 07/26
University of Nantes France
 - Level: Very good distinction.
 - Main courses: Remote Sensing/GIS, Surface Processes, Data Analysis.
- **Earth Sciences' Bachelor** 08/24
University of La Rochelle France
 - Level: Valedictorian and very good distinction.
 - Erasmus exchange at the University of Helsinki.
 - Advanced courses in Engineering and Research Initiation.
- **Esabac High School Diploma** 08/21
Jean Perrin High School Lyon
 - Level: Good distinction.
 - Binational french and italian path.

INTERNSHIPS

- **Super-resolved infrared mapping of Titan** 01/26 - 06/26
Institut de Physique du Globe de Paris Paris
 - Under the supervision of Dr. Sébastien RODRIGUEZ
 - Deployment of a super-resolution algorithm on the Cassini/VIMS hyperspectral dataset on Titan.
 - Leading a spectral and geomorphological study of some regions of interest.
 - Perform photometric correction of infrared observations by Cassini/VIMS.
- **Detection of dark spots from seasonal sublimation of CO_2 on Mars** 06/25 - 08/25
Laboratoire de Planétologie et Géosciences, Center for Space and Habitability Nantes, Bern
 - Under the supervision of Dr. Clémence HERNY, Dr. Valentin BICKEL & Dr. Susan CONWAY.
 - Training and deployment of a deep learning model to identify dark spots features on Mars, related to the sublimation of CO_2 ice.
 - Statistical study of the abundance and density trends of dark spots observed by MRO/CTX.
- **Characterization of Enceladus' terrains through multi-wavelength observations** 04/25 - 06/25
Laboratoire Atmosphère et Observations Spatiales Paris
 - Under the supervision of Dr. Alice LE GALL & Salman RAZA.
 - Provide an estimation range of penetration depth of radar waves into Enceladus' icy shell by Cassini/SAR.
 - Constrain properties of surface and subsurface of Enceladus' terrains, thanks to scatterometric analysis.
 - Propose a geological mapping in South Polar Terrains.

- **Preliminary study of synergies between JUICE and Europa Clipper missions** 06/24 - 07/24
Laboratoire de Planétologie et Géosciences Nantes
 - Under the supervision of Dr. Gabriel TOBIE & Dr. Benoît SEIGNOVERT.
 - Identification of geologically Regions of Interest expected to be seen by both missions.
 - Propose a scenario for data acquisition of JUICE/MAJIS, JUICE/JANUS, Europa Clipper/SUDA, Europa Clipper/MISE and Europa Clipper/EIS.
- **Quantitative estimation of environmental impact of plastic in coastal areas** 05/22 - 06/22
Laboratoire Littoral Environnement et Sociétés La Rochelle
 - Under the supervision of Dr. Jean-Michel CAROZZA.
 - Elaboration of an inventory of plastic materials found on the field (foreshore of La Rochelle).
 - Meeting and presentation of results to local actors (politics, oyster farmers...).

PUBLICATIONS

Raza S., Le Gall A., Schmidt F., Benedikter A., Stiles B.W., Picard G., Rosen P.A., Mergny C. & **Vignon P.A.** (2026). Joint analysis of the Cassini RADAR Active and Passive Observations of Enceladus: Constraints on Regolith Properties and Subsurface Thermal Structure in the South Polar Terrain. *Icarus* (submitted).

Vignon P.A., Bickel V.T., Herny C., & Conway S.J. (2026). Dark spots as spatiotemporal indicators of sublimation of CO₂ seasonal polar caps on Mars. *Geophysical Research Letters* (in preparation).

Vignon P.A., Bickel V.T., Herny C., Conway S.J. (2026). *Dark spots on Mars*. [Dataset]. BORIS University of Bern (in preparation).

PROFESSIONAL AND VOLUNTEER EXPERIENCE

- **Private Teacher of Mathematics** Since 09/22
Superprof, Completude La Rochelle, Nantes, Paris
 - Teaching Mathematics and Sciences to high school students.
- **Member of the Council for Development** 09/22 - 06/24
La Rochelle's agglomeration La Rochelle
 - Propose projects about transports, energies and environmental policies at the scale of the metropolitan area.
 - Give feedback about policies as a citizen.
- **Founding member and general secretary** 09/21 - 12/22
Egalit'R La Rochelle
 - Founded and managed a student association.
 - Organize awareness days in high schools.
 - Organize cleanwalks in cities and on beaches.

SKILLS

- **Programming Languages:** Python (Data Analysis, deep learning/CNN, numpy, matplotlib, pandas, scipy, scikit-learn, seaborn), basis in IDL.
- **Geographic Information Systems:** QGIS, SQL, Python.
- **Image and text treatment:** Inkscape, Office (Excel, Word...), LaTeX.
- **Languages:** French (mother tongue), English (C1), Italian (B2), Finnish (A1).

CERTIFICATIONS

- **TOEIC** English proficiency (910/990). 01/26
- **Voltaire** French orthographic skills and knowledge. 12/22
- **ESABAC** Italian proficiency (94/100). 06/21
- **PIX** Ability to handle Office (Excel, Word...). 05/22
- **Boating licences** River, Coastal, and Deep sea. 2023-24
- **Driving licence.** 09/22